

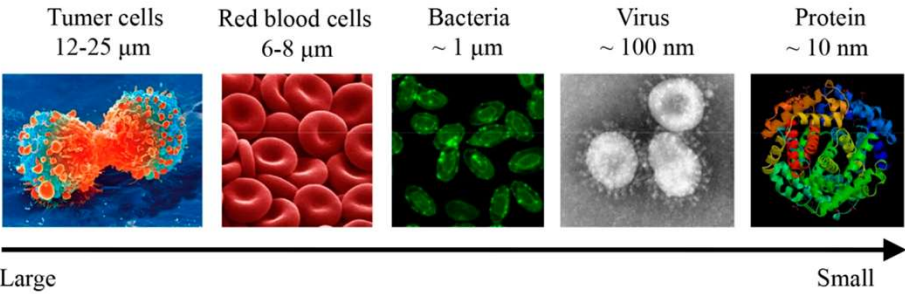
Índice del tema ix

► ix. Fluidos complejos en chips

- Definiciones y problemática
- Sangre en microcanales
 - Interacción con la pared
 - Carácter no-Newtoniano
- Aplicaciones en *lab-on-a-chips*

ix. Fluidos complejos

micro/nano partículas
(bio) macromoléculas

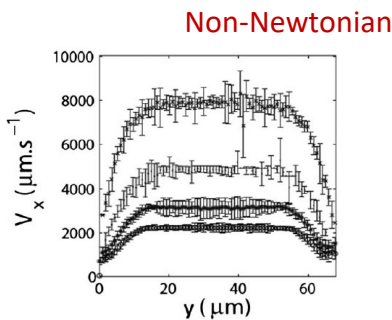
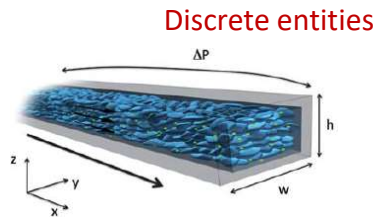
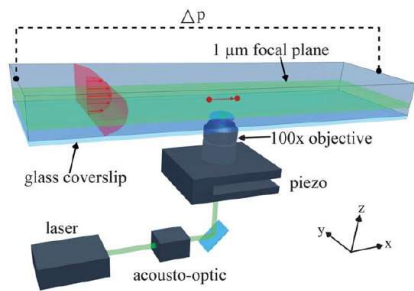


Liu and Hu, *Micromachines* **8** (2017) 73

ix. Microfluídica para fluidos complejos

Microfluidics and complex fluids†

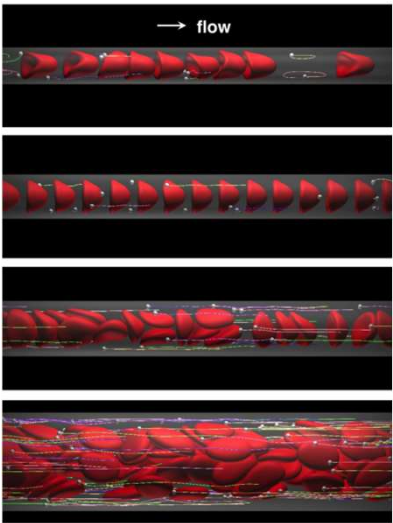
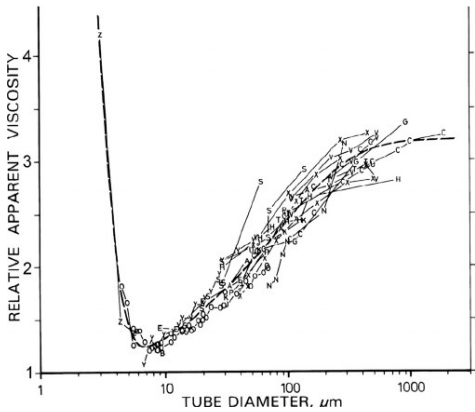
Ph Nghe,^a E. Terriac,^a M. Schneider,^a Z. Z. Li,^a
M. Cloitre,^b B. Abecassis^a and P. Tabeling^{ab}



ix. Fluidos complejos



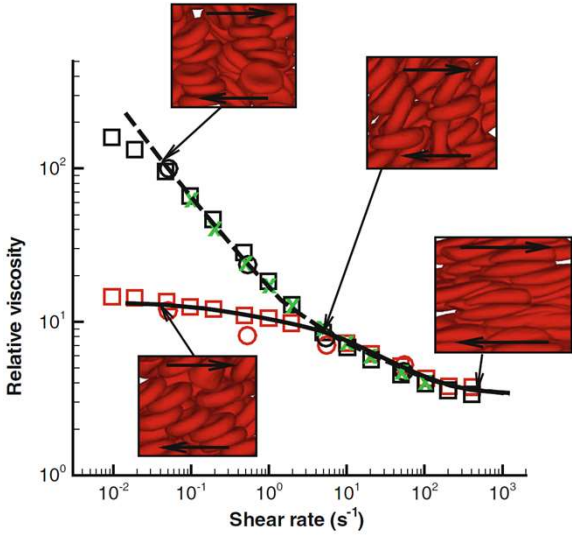
ix. Sangre: efecto del tamaño de canal



Fåhræus R. and Lindqvist T.
The viscosity of the blood in
narrow capillary tubes.
Am J Physiol **96** (1931) 562–568

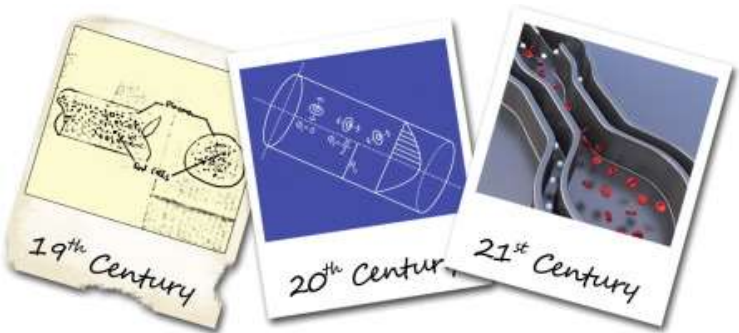
Takeishi and Imai,
Scientific Reports **7** (2017) 5381

ix. Sangre: efecto de la velocidad de corte



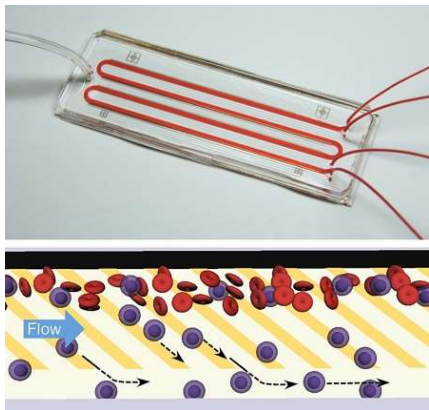
Fedosov et al, *Biomech Model Mechanobiol* **13** (2014) 239

ix. Sangre en chips

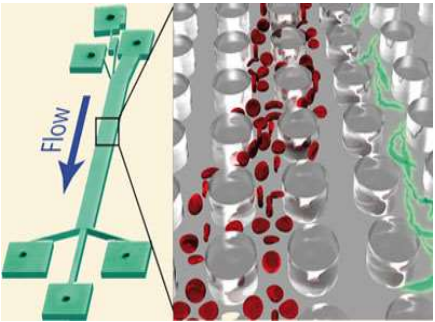


Geislinger and Franke, *Adv Coll Int Sci* (2014)

ix. Sangre en chips



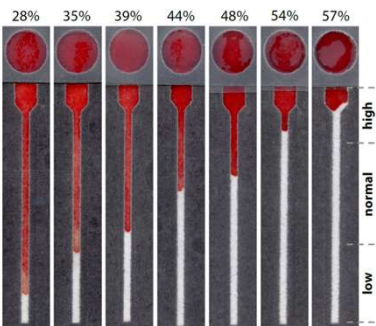
news.mit.edu/2013/research-update-new-microchip-sorts-white-blood-cells-from-whole-blood-0806



rsc.org/chemistryworld/News/2011/March/22031101.asp

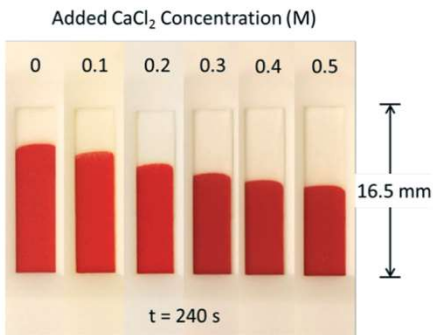
ix. Sangre en chips

Hematócrito:



Berry et al, *Lab Chip* **16** (2016)

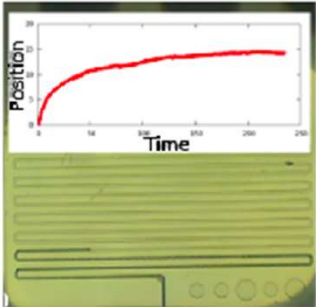
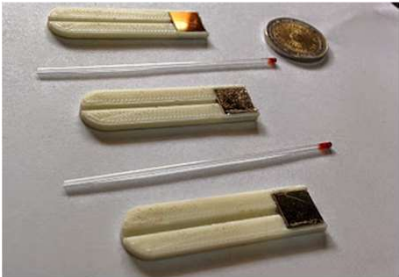
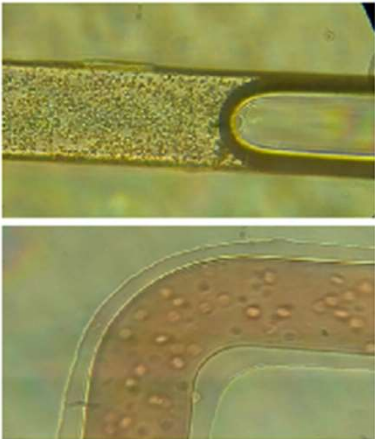
Coagulación:



Li et al, *Lab Chip* **14** (2014)

ix. Sangre en chips

Viscosidad:



Morhell and Pastoriza, *Sensors and Actuators B* **227** (2016) 24